Ohio River Information

The Ohio River begins at the confluence of the Allegheny and Monongahela rivers at the Point in Pittsburgh, PA, and flows 981 miles to join the Mississippi at Cairo, Ill.

The importance of this great river for navigation and trade was recognized first by the 17th century European imperial powers in North America. France claimed the territory drained by La Belle Riviere on the basis of explorations made by La Salle in 1669. England later claimed the same land by a purchase from Native Americans in 1744. Conflict over their colonial possessions drew the two countries into the French and Indian War, which lasted from 1756 to 1763.

The English victory in the war cleared the way for westward expansion from the English colonies of the eastern seaboard, and thousands of settlers began moving into the Ohio country. This immigration was slowed by the American Revolution, but after the war, the great migration into the western lands continued.

In that era of primitive transportation, the Allegheny Mountains posed the greatest barrier to westward expansion. The two principal routes were overland from Baltimore to Redstone on the Monongahela River via the National Road; or by the Forbes Road from Philadelphia to Pittsburgh. At the end of these two overland treks, the settlers bought or constructed boats and rafts and continued their journey by water.

The flatboat was the cheapest of the many types of boats used and became the standard conveyance for families moving west. All of the boats in this period were hand-powered, with poles or oars for steering, and usually floated with the current. They were not intended for round trips since the settlers used them only to get to their new homes and then broken them up for their lumber.

This situation changed dramatically in 1811 with the launching of the first steamboat on the western waters, the New Orleans, which was built near Pittsburgh. Steamboats made it possible to increase the speed of the trip downriver and made the return trip easier. Commerce on the rivers increased and by the end of 1835 more than 650 steamboats had been built in the west, including 304 in the vicinity of Pittsburgh.

However, the conditions of the rivers made navigation difficult. Shifting sand bars, snags and rocks combined with seasonally fluctuating river depths made river travel dangerous. Mark Twain has immortalized the era of the river pilots who were required to memorize every foot of the river in order to steer the steamboats safely through the many hazards. Even so, boats were wrecked and the increasing amount of trade on the rivers made navigating safely on them of primary importance. River users demanded the federal government step in and improve the rivers.

The first steps were taken in 1824, with an act of Congress authorizing the removal of snags and sandbars from the Mississippi and Ohio rivers. The U.S. Army Corps of Engineers was put in charge of this work.

In the year following, hazards were removed from the river and travel became safer, but the problem of low water remained. State and local governments and private companies attempted to solve this problem but they lacked the resources or the jurisdiction to undertake the massive project.

Again Congress acted, and in 1878, the U.S. Army Corps of Engineers began the first federally -built lock and dam on the Ohio at Davis Island about five miles below the Point in Pittsburgh. This lock and dam was completed in 1885. The lock was 110 feet wide and 600 feet long and was the largest lock in the world at that time. The dam was composed of wooden bulkheads hinged on the river bottom which could be lowered when the river flow was high. Boats could then pass without using the lock. When the level of the river began to fall, however, the wooden wickets were raised to catch the water and create a pool behind the dam to maintain the level of the river for boats. These wicket dams were eventually built for the entire length of the Ohio River from Pittsburgh to Cairo, Illinois. The last one, Dam 53 at Cairo, was installed in 1929. It and Dam 52 are the only wicket dams remaining on the Ohio River.

Commercial traffic on the Ohio River has increased over the years with the growth of heavy industry in the tri-state area. In the first year that tonnage records were kept by the Corps (1917), the Ohio River carried about 5 million tons of cargo. Now, commerce is approaching 150 million tons a year. The increase in traffic over the years has been met by improvements and modernization of the navigation facilities.

As the tows on the rivers became larger, the wicket dams with their small lock chambers were inadequate and new facilities were built. Even before the wicket dam system was completed in 1929, some of the old structures were being replaced. The original Davis Island Lock and Dam was removed from the river and replaced in 1922.

The Pittsburgh District now has six modern lock and dams on the Ohio River where once 14 wicket dams were needed. The last of the old dams was removed from the river in 1975 with the completion of Hannibal Lock and Dam.

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